GRAND ROUNDS CASE · FOLLOW-UP PROGRAM

# Followup 26-15

Respiratory arrest progressing to cardiac arrest

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|---|---|
| **Patient** | A 52-year-old female, a former smoker with no known medical history, obese, with a weight of 129 kg recorded for drug dosing. Her family reported a viral upper respiratory infection the day before and shortness of breath before the call. **She was found in respiratory arrest with a pulse.** |
| **Outcome** | She lost her pulse about five minutes after BLS contact, in asystole, and stayed in cardiac arrest through EMS care. In the emergency department she had a brief return of circulation, then arrested again. **She died in the emergency department. The hospital recorded respiratory arrest and cardiac arrest and did not determine a cause.** |
| **Disposition** | Died in the emergency department after roughly 100 minutes of total resuscitation, with her husband and two children at the bedside. The case was referred to the medical examiner. |

## The call

The call was dispatched as difficulty breathing. Two of her sons met the BLS crew at the front porch and told them she had collapsed and was not waking up. The family said she had told them she was having a very hard time breathing and to call 911.

The BLS crew found her unconscious and unresponsive, with slow, labored, gasping respirations at a rate of 2 and a palpable radial pulse. They started bag-valve-mask ventilation with oxygen at 15 LPM within seconds of contact. The Echo paramedic arrived about a minute later and found carotid and radial pulses, skin that was pink, warm and dry, pupils equal and reactive, no sign of injury, and no lung sounds on either side. Epinephrine 0.3 mg IM was given for respiratory distress with poor air movement.

She then lost her pulse. The monitor showed asystole, compressions began, and mechanical CPR took over within about two minutes. A BLS provider placed an i-gel, the paramedic student placed a left tibial IO, and an 18 gauge IV went into the right antecubital fossa. Her glucose was 215. She received five doses of epinephrine and was transported with mechanical CPR running. She had no return of circulation during EMS care.

Times below are from BLS patient contact. Where the cardiac monitor recorded an event, its time is used; the BLS and Echo clocks agreed within seconds.

| From contact | Finding or action |
|---|---|
| 0 | Unresponsive, gasping at 2 per minute, radial pulse present |
| Under 1 min | Bag-valve-mask ventilation with oxygen at 15 LPM |
| +1 min | Echo paramedic at bedside: carotid and radial pulses, no lung sounds bilaterally; epinephrine 0.3 mg IM |
| +3 min | Cardiac monitor on |
| +5 min | Pads on; asystole |
| +6 min | Compressions begin |

| From contact | Finding or action |
|---|---|
| +7 min | Mechanical CPR running |
| +13 min | First end-tidal CO2 through the i-gel: 90 mmHg |
| +15 to +28 min | Epinephrine 1 mg five times, three IO and two IV |
| +38 min | Leaves the scene with mechanical CPR running |
| +41 min | Arrives at the emergency department |

The numbers to hold onto are about five minutes with a pulse, and a first end-tidal CO2 of 90.

## Primary impression

The BLS crew documented respiratory arrest with a pulse. The hospital's two final diagnoses were cardiac arrest and respiratory arrest. This crew recognized that a woman with a radial pulse and two gasps a minute was not breathing, and they ventilated her within seconds. They had it right.

The mechanism underneath is hypoxic cardiac arrest. Her heart did not stop first. Her breathing failed, her blood ran out of oxygen, and her heart stopped because it was starved of it. What caused the respiratory failure was never established. The CT scans ordered in the emergency department never resulted, and the case went to the medical examiner.

## Hospital course

**Emergency department, day 0.** She arrived in asystole with mechanical CPR running and the i-gel in place. She was intubated on the first attempt by video laryngoscopy with an 8.0 tube, confirmed by waveform capnography, colorimetric CO2, direct visualization, and chest x-ray. The team documented difficulty bagging her and end-tidal CO2 readings that swung between the 30s and the 80s. A venous blood gas showed acidosis, and she received four doses of sodium bicarbonate, calcium chloride, six more doses of epinephrine, and nebulized albuterol.

Her rhythm moved from asystole to PEA. She had a brief return of spontaneous circulation, and vasopressin and norepinephrine infusions were started. She lost her pulse again, developed ventricular tachycardia, and was defibrillated twice at 200 J. Bedside cardiac ultrasound at the end of the resuscitation showed no cardiac activity and no pericardial effusion. Her husband and two children were at the bedside. Resuscitation was stopped after about an hour in the department, and the medical examiner was notified.

**Final diagnoses:** Cardiac arrest. Respiratory arrest.

## Education points

### Hypoxic cardiac arrest

Let's start with why breathing failure ends in cardiac arrest, and how fast. Oxygen is stored in three places: the air left in the lungs at the end of a breath, hemoglobin, and a small amount dissolved in plasma. When breathing stops, the body keeps consuming oxygen at about 250 mL a minute, and those stores run down within minutes.

The air left in the lungs at the end of a normal breath is called the functional residual capacity. It is the largest of the three stores and the one that buys time. Obesity shrinks it, because the weight of the chest wall and

abdomen compresses the lungs, and it raises oxygen consumption at the same time. A patient with her body habitus desaturates far faster than a lean patient once breathing fails.

As the oxygen falls, the heart responds in a recognizable order. It first speeds up from the catecholamine surge. Then the myocardium itself becomes hypoxic, the conduction system slows, and the rhythm drifts toward bradycardia. The ventricles stop contracting effectively while electrical activity continues, which is PEA, and finally the electrical activity stops, which is asystole. That is why hypoxic arrests almost always present in asystole or PEA, and almost never in ventricular fibrillation.

Gasping at a rate of 2 is not breathing. It is a brainstem reflex that persists after effective ventilation has stopped, and it moves almost no air. A patient who is gasping with a pulse is in respiratory arrest, and the window before the heart stops is the only time ventilation alone can reverse the process. Once the heart has stopped, restoring oxygen no longer restores the pump.

She had about five minutes between BLS contact and the loss of her pulse. The ventilation started in that window was the correct intervention.

## Ventilating against airflow obstruction

The hospital did not name a cause, and this document does not either. Several findings in the record are consistent with severe airflow obstruction in the lower airways: a viral illness the day before, shortness of breath, poor air movement and no lung sounds before the arrest, difficulty bagging in the emergency department, very high end-tidal CO2, and nebulized albuterol given by the hospital team. Severe bronchospasm is the most common cause of that picture. An airway obstruction, aspiration, and pulmonary embolism were not distinguished by any test that resulted. A large pulmonary embolism usually lowers end-tidal CO2, because blood stops reaching ventilated lung, so readings as high as hers argue against it without excluding it.

Absent lung sounds in a patient struggling to breathe are called a silent chest. It means so little air is moving that the airways no longer produce sound. In severe bronchospasm it is a sign of near-complete obstruction, not improvement.

Lungs with narrowed airways fill easily and empty slowly. Inhalation is active and can force air past the narrowing, but exhalation is passive and depends on time. If the next breath arrives before the last one has left, air accumulates breath by breath. This is called dynamic hyperinflation, and the pressure left in the chest at the end of each exhalation is called auto-PEEP.

Auto-PEEP matters to circulation as much as to breathing. Rising pressure in the chest compresses the great veins, venous return falls, and cardiac output falls with it. Enough of it can produce hypotension or PEA in a patient whose heart is otherwise able to beat, and it makes compressions less effective during an arrest. The bag gets harder to squeeze with each breath, which matches the difficulty the emergency department documented.

The ventilation strategy that follows from this physiology applies to any patient with airflow obstruction. Use a slow rate, a modest volume, and a long pause to let the chest fall fully before the next breath. If the patient becomes hard to bag and loses perfusion, disconnecting the bag for several seconds lets trapped air escape, and firm pressure on the chest can speed it. If perfusion returns after a disconnect, auto-PEEP was the problem.

## End-tidal CO2 after an asphyxial arrest

During CPR for a cardiac arrest that starts in the heart, such as ventricular fibrillation, end-tidal CO2 measures blood flow. Carbon dioxide reaches the lungs only as fast as compressions push blood through them, so a higher reading means better perfusion, and a sudden rise suggests return of circulation.

An arrest that starts with breathing failure is different. For minutes before the heart stops, the body keeps producing carbon dioxide and has no way to get rid of it. The blood and tissues fill with it. When ventilation and compressions begin, that stored carbon dioxide arrives at the lungs all at once, and the first readings are high for reasons that have nothing to do with the quality of the CPR.

Prehospital studies have measured this. In one study, the first end-tidal CO2 in asphyxial arrests averaged about 50 mmHg, against about 34 in arrests from ventricular fibrillation. In the asphyxial group, the first reading did not separate patients who regained a pulse from those who did not. After about five minutes of CPR, the readings in both groups began to separate survivors from non-survivors.

Her first reading through the i-gel was 90 mmHg, then 86. Over the next half hour her readings fell into the 50s and 60s, with one at 26, and finished at 59. The early values reflect carbon dioxide accumulated during respiratory failure. The trend after the stored carbon dioxide has been washed out is the more useful number for judging perfusion.

> **Further reading on end-tidal CO2 in asphyxial arrest.** The full prehospital study comparing asphyxial and ventricular fibrillation arrests, with the minute-by-minute pattern of end-tidal CO2 in each. https://pmc.ncbi.nlm.nih.gov/articles/PMC3222045/
>
> **Further reading on ventilating severe bronchospasm.** The EMCrit Internet Book of Critical Care chapter on critical asthma, including auto-PEEP and how it causes circulatory collapse. https://emcrit.org/ibcc/asthma/

### Prehospital considerations

**A pulse with no effective breathing is a ventilation emergency.** The BLS crew found a radial pulse and could have read her as unconscious but stable. They read the gasping correctly and were ventilating within seconds. Ventilation before the arrest was the only intervention with a chance of preventing it.

**Obesity shortens the time to desaturation and makes ventilation harder.** A reduced functional residual capacity means less stored oxygen, and a heavy chest wall resists the bag. Two-handed mask seal with a second provider squeezing, and early placement of a supraglottic airway, are the tools for that problem.

**Mechanical CPR keeps compression quality through long resuscitations and patient movement.** The monitor record shows compressions running 87 percent of the time from the first compression to the hand-off, about 3,150 compressions at a steady 102 per minute, through IO and IV access, five medication doses, packaging and transport. That is good resuscitation work by everyone on the call.

## The interventions

#### PREHOSPITAL INTERVENTIONS

**Bag-valve-mask ventilation with oxygen at 15 LPM.** Started within seconds of contact in a patient with a pulse and agonal respirations. This was the treatment for the problem she had, and it was in place before anything else.

**Supraglottic airway, i-gel, placed by a BLS provider.** Placed during the arrest and confirmed three ways: lung sounds, absent epigastric sounds, and waveform capnography. Waveform capnography is the confirmation that carries the weight, because a sustained waveform with every breath cannot be produced by a device that is not ventilating the lungs. Her first reading of 90 mmHg proved the i-gel was moving gas in and out of the lungs. The emergency department found it in place and working, and intubated on the first attempt. A BLS provider placing an airway that held through the rest of the resuscitation, mechanical CPR and transport is exactly the airway management this program wants to see.

**Epinephrine 0.3 mg IM.** Given before the arrest for respiratory distress with poor air movement. Epinephrine relaxes bronchial smooth muscle through beta-2 receptors and tightens leaking vessels through alpha-1 receptors, which is why it is the first drug for anaphylaxis and is used in severe bronchospasm. An IM dose depends on blood flow to the muscle to reach the circulation, and that flow falls as a patient approaches arrest.

**Mechanical CPR.** Applied within about two minutes of the first compression and run continuously through transport.

**Intraosseous access, left tibia, placed by the paramedic student, and an 18 gauge IV in the right antecubital fossa.** Both were used for the epinephrine doses. IO access in a patient this size is not trivial, and it worked.

**Epinephrine 1 mg IV and IO, five doses.** Given at intervals of 3 to 4.5 minutes. In asystole and PEA, epinephrine raises aortic diastolic pressure through alpha-1 vasoconstriction, which raises coronary perfusion pressure during compressions and increases the chance of a return of circulation.

#### IN-HOSPITAL INTERVENTIONS

**Endotracheal intubation.** First attempt by video laryngoscopy during CPR, exchanging a working supraglottic airway for a cuffed tube.

**Sodium bicarbonate and calcium chloride.** Four doses of bicarbonate for acidosis on the blood gas, and calcium. Bicarbonate generates carbon dioxide as it buffers acid, which contributes to the swings in end-tidal CO2 the team documented.

**Bedside cardiac ultrasound.** Repeated during the resuscitation. At the end it showed no cardiac activity and no pericardial effusion, which excluded tamponade as a reversible cause.

## The labs, and what they mean

*Reference ranges vary between institutions and analyzers. These are the ranges printed on the source reports.*

### Metabolic consequences of prolonged arrest

| Lab | Value | Reference range |
|---|---|---|
| Lactic acid (mmol/L) | 22.0 | 0.0 to 2.0 |
| Phosphorus (mg/dL) | 12.1 | 2.5 to 4.5 |
| Magnesium (mg/dL) | 3.2 | 1.6 to 2.5 |

**A lactate of 22 is eleven times the top of the range.** Cells without oxygen switch to anaerobic metabolism, which produces lactate and acid. This value reflects the minutes of hypoxia before the arrest and the low flow of a long resuscitation. Phosphorus and magnesium are mostly held inside cells, and both rise when cells are injured and when acidosis shifts them out.

### Blood count and toxicology

| Lab | Value | Reference range |
|---|---|---|
| Hemoglobin (g/dL) | 10.5 | 12.0 to 16.0 |
| Mean cell hemoglobin (pg) | 25.2 | 27.0 to 33.0 |

| Lab | Value | Reference range |
|---|---|---|
| Mean cell hemoglobin concentration (g/dL) | 26.6 | 32.0 to 37.0 |
| Lymphocytes, absolute (x10^9/L) | 7.22 | 1.00 to 4.00 |
| Neutrophils, absolute (x10^9/L) | 1.62 | 1.80 to 7.80 |

Ethanol and salicylate levels were normal, acetaminophen was undetectable, and pro-BNP was normal. Troponin, the metabolic panel, the blood gas values and the urine drug screen do not appear in the record.

**Two values worth reading on your own.** Her absolute lymphocyte count was 7.22, nearly twice the top of the range, with neutrophils slightly low. Viral infections raise lymphocyte counts, which fits the illness her family described. Acute stress does too: catecholamines, including the epinephrine given during a resuscitation, release lymphocytes into the circulation within minutes. A single sample drawn during cardiac arrest cannot separate the two.

Her hemoglobin was 10.5, and the red cells were small and pale: a low mean cell hemoglobin and a low mean cell hemoglobin concentration. That pattern is called microcytic, hypochromic anemia, and iron deficiency is its most common cause. It develops over months and did not cause this event. A lower hemoglobin does carry less oxygen, which shortens the time to severe hypoxia once breathing fails.

## Learning points

1. **A patient who is gasping with a pulse is in respiratory arrest, and ventilation is the treatment.** Agonal respirations move almost no air. This crew read a rate of 2 with a radial pulse correctly and ventilated within seconds, which is the only intervention that can stop a hypoxic arrest before it happens.
2. **Hypoxic cardiac arrest comes on fast, faster in obesity.** Breathing failure drains the oxygen stores within minutes, the heart slows, and the arrest arrives in PEA or asystole. A reduced functional residual capacity and higher oxygen consumption shorten that window. She had about five minutes.
3. **Obstructed lungs need time to empty.** Ventilating a patient with severe airflow obstruction at a normal rate traps air, raises the pressure in the chest, and cuts venous return. A slow rate and a long exhalation protect circulation, and a brief disconnect is the test and the treatment when perfusion falls.
4. **The first end-tidal CO2 after an asphyxial arrest measures stored carbon dioxide, not perfusion.** Her first reading was 90 mmHg. In an arrest that starts with breathing failure, the trend after several minutes of CPR is the number that reflects blood flow.
5. **A supraglottic airway confirmed by waveform capnography is a working airway.** A BLS provider placed an i-gel that ventilated her through mechanical CPR, transport and hand-off, and waveform capnography proved it from the first breath.

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*Prepared from records obtained through the follow-up request process and de-identified for education. To request follow-up on a call you ran, fill out the request form at [https://www.berrienmca.org/followup/](https://www.berrienmca.org/followup/)*